

S.5

$$(a) y_i = B_1 + B_2 t + e_i, \quad t = 1, 2, \dots, T$$

$$\begin{aligned} \sum_{i=1}^T (t - \bar{t})^2 &= \sum_{i=1}^T t^2 - \left(\sum_{i=1}^T t\right)^2 / T \\ &= \frac{T(T+1)(2T+1)}{6} - \frac{[T(T+1)/2]^2}{T} = \frac{T(T+1)(2T+1)}{6} - \frac{T(T+1)^2}{4} = \frac{T(T^2-1)}{12} \end{aligned}$$

$$\text{故 } \text{Var}(b_2 | T) = \frac{\sigma^2}{\frac{T(T^2-1)}{12}} = \frac{12\sigma^2}{T(T^2-1)} = \frac{12\sigma^2}{T^3-T} \rightarrow 0$$

$$\lim_{T \rightarrow \infty} \text{Var}(b_2 | T) = \lim_{T \rightarrow \infty} \frac{12\sigma^2}{T^3-T} = 0$$

$$\begin{cases} \lim_{T \rightarrow \infty} E(b_2 | X) = B_2 \\ \lim_{T \rightarrow \infty} \text{Var}(b_2 | T) = 0 \end{cases} \Rightarrow \text{估計量 } b_2 \text{ 滿足一致性} \quad \#$$

$$(b) Y_t = B_1 + B_2 (0.5)^t + e_t, \quad \text{independent variable } x_t = 0.5^t$$

$$\sum_{t=1}^T (0.5^t - \bar{x})^2 < \sum_{t=1}^T (0.5^t)^2 = \sum_{t=1}^T 0.25^t = \frac{0.25}{1-0.25} = \frac{1}{3}$$

由於 $\sum (x_i - \bar{x})^2$ 收斂到一個有限常數 ($< \frac{1}{3}$), 非發散到無窮大

$$\text{故 } \lim_{T \rightarrow \infty} \text{Var}(b_2 | X) = \lim_{T \rightarrow \infty} \frac{\sigma^2}{\sum (0.5^t - \bar{x})^2} = 3\sigma^2 \neq 0$$

表示估計量 b_2 不符合一致性。#

(c)

(a) 小題隨著T增加，時間T會不斷變大，代表每一次增加樣本，都在觀察更極端的X值，提供了關於斜率B2的全新且強大的資訊，因此能越來越準確地抓住趨勢。

(b) 小題隨著T增加， 0.5^t 會消失得非常快 (0.5, 0.25, 0.125, ...)。當T很大以後，後面的項幾乎都等於 0，代表增加再多的樣本，也只是在重複收集接近0的資訊，並沒有提供新的變動來幫助修正對B2的估計，所以誤差永遠無法消失。

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5-6

$$\begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \\ -1 \end{bmatrix}, \quad \widehat{Cov}(b_1, b_2, b_3) = \begin{bmatrix} 3 & -2 & 1 \\ -2 & 4 & 0 \\ 1 & 0 & 3 \end{bmatrix}$$

(a)

$$H_0: B_2 = 0 \quad \text{vs} \quad H_1: B_2 \neq 0$$
$$b_2 \sim N(B_2, \text{Var}(b_2))$$

test statistic

$$T = \frac{b_2 - 0}{\text{se}(b_2)} \sim t(63-3=60)$$

$$\text{Rejection Region: } \{t^* \mid |t^*| > t_{0.025}(60) = 2.00\}$$

$$\therefore T = \frac{3-0}{\sqrt{4}} = 1.5 \notin \text{RR} \quad \therefore \text{Do not reject } H_0 \quad \alpha = 0.05$$

There isn't enough evidence to say that $B_2 \neq 0$ #

(b) $H_0: \beta_1 + 2\beta_2 = 5$ vs $H_1: \beta_1 + 2\beta_2 \neq 5$

$$b_1 + 2b_2 \sim N(\beta_1 + 2\beta_2, \text{Var}(b_1 + 2b_2))$$

$$\begin{aligned}\text{Var}(b_1 + 2b_2) &= \text{Var}(b_1) + 4\text{Var}(b_2) + 4 \cdot \text{cov}(b_1, b_2) \\ &= 3 + 4 \cdot 4 + 4 \cdot (-2) \\ &= 11\end{aligned}$$

$$\text{se}(b_1 + 2b_2) = \sqrt{\text{Var}(b_1 + 2b_2)} = \sqrt{11}$$

$$b_1 + 2b_2 = 2 + 2 \cdot 3 = 8$$

Test statistic

$$T = \frac{b_1 + 2b_2 - 5}{\text{se}(b_1 + 2b_2)} \sim t(60)$$

$$RR: \{t^* \mid |t^*| > t_{0.025}(60) = 2.00\}$$

$$\therefore T = \frac{8-5}{\sqrt{11}} = 0.9045 \notin RR \quad \therefore \text{Do not reject } H_0 \text{ at } \alpha = 0.05$$

There isn't enough evidence to say that $\beta_1 + 2\beta_2 \neq 5$ #

$$(c) H_0: B_1 - B_2 + B_3 = 4 \quad \text{vs} \quad H_1: B_1 - B_2 + B_3 \neq 4$$

$$b_1 - b_2 + b_3 \sim N(B_1 - B_2 + B_3, \text{Var}(b_1 - b_2 + b_3))$$

$$b_1 - b_2 + b_3 = 2 - 3 + (-1) = -2$$

$$\begin{aligned} \text{Var}(b_1 - b_2 + b_3) &= \text{Var}(b_1) + \text{Var}(b_2) + \text{Var}(b_3) - 2\text{cov}(b_1, b_2) + 2\text{cov}(b_1, b_3) - 2\text{cov}(b_2, b_3) \\ &= 3 + 4 + 3 - 2 \cdot (-2) + 2 \cdot 1 - 2 \cdot 0 \\ &= 16 \end{aligned}$$

$$\text{se}(b_1 - b_2 + b_3) = \sqrt{16} = 4$$

Test statistic

$$T = \frac{b_1 - b_2 + b_3 - 4}{\text{se}(b_1 - b_2 + b_3)} \sim t(60)$$

$$RR: \{ t^* \mid |t^*| > t_{0.025}(60) = 2.00 \}$$

$$\therefore T = \frac{-2-4}{4} = -1.5 \notin RR \quad \therefore \text{Do not reject } H_0 \text{ at } \alpha = 0.05$$

There isn't enough evidence to say that $B_1 - B_2 + B_3 \neq 4$ #

5-20

$$(a) \hat{B}_{2, \text{mean}} = \frac{\bar{y}_2 - \bar{y}_1}{\bar{x}_2 - \bar{x}_1} = \frac{(B_1 + B_2 \bar{x}_2 + \bar{e}_2) - (B_1 + B_2 \bar{x}_1 + \bar{e}_1)}{\bar{x}_2 - \bar{x}_1} = B_2 + \frac{\bar{e}_2 - \bar{e}_1}{\bar{x}_2 - \bar{x}_1}$$

$$\begin{aligned} E(\hat{B}_{2, \text{mean}}) &= E\left(B_2 + \frac{\bar{e}_2 - \bar{e}_1}{\bar{x}_2 - \bar{x}_1}\right) = B_2 + \frac{1}{\bar{x}_2 - \bar{x}_1} \cdot [E(\bar{e}_2 | X) - E(\bar{e}_1 | X)] \\ &= B_2 + 0 \\ &= B_2 \end{aligned}$$

$\therefore E(\hat{B}_{2, \text{mean}}) = B_2 \quad \therefore \hat{B}_{2, \text{mean}}$ is unbiased #

$$(b) \text{Var}\left(\frac{\bar{e}_2}{\bar{x}_2 - \bar{x}_1}\right) = \frac{1}{(\bar{x}_2 - \bar{x}_1)^2} \cdot \text{Var}(\bar{e}_2) = \frac{1}{(\bar{x}_2 - \bar{x}_1)^2} \cdot \frac{\sigma^2}{N/2} = \frac{2\sigma^2}{N(\bar{x}_2 - \bar{x}_1)^2}$$

$$\text{同理 } \text{Var}\left(\frac{\bar{e}_1}{\bar{x}_2 - \bar{x}_1}\right) = \frac{2\sigma^2}{N(\bar{x}_2 - \bar{x}_1)^2}, \quad B_2 \text{ is constant}$$

$$\begin{aligned} \text{Var}(\hat{B}_{2, \text{mean}}) &= \text{Var}\left(B_2 + \frac{\bar{e}_2 - \bar{e}_1}{\bar{x}_2 - \bar{x}_1}\right) = \text{Var}\left(\frac{\bar{e}_2 - \bar{e}_1}{\bar{x}_2 - \bar{x}_1}\right) \\ &= \text{Var}\left(\frac{\bar{e}_2}{\bar{x}_2 - \bar{x}_1}\right) + \text{Var}\left(\frac{\bar{e}_1}{\bar{x}_2 - \bar{x}_1}\right) \\ &= \frac{2\sigma^2}{N(\bar{x}_2 - \bar{x}_1)^2} + \frac{2\sigma^2}{N(\bar{x}_2 - \bar{x}_1)^2} = \frac{4\sigma^2}{N(\bar{x}_2 - \bar{x}_1)^2} \quad \# \end{aligned}$$

$$\begin{aligned} (c) \text{Var}(\hat{B}_{2, \text{mean}}) &= \text{Var}(E[\hat{B}_{2, \text{mean}} | X]) + E[\text{Var}(\hat{B}_{2, \text{mean}} | X)] \\ &= \text{Var}(B_2) + E\left(\frac{4\sigma^2}{N(\bar{x}_2 - \bar{x}_1)^2}\right) \\ &= E\left[\frac{4\sigma^2}{N(\bar{x}_2 - \bar{x}_1)^2}\right] = \frac{4\sigma^2}{N} \cdot E\left[\frac{1}{(\bar{x}_2 - \bar{x}_1)^2}\right] \quad \# \end{aligned}$$

(d)

$$\begin{cases} \lim_{N \rightarrow \infty} E(\hat{B}_{2, \text{mean}}) = B_2 \\ \lim_{N \rightarrow \infty} \text{Var}(\hat{B}_{2, \text{mean}}) = 0 \end{cases}, \text{ 此時 } \hat{B}_{2, \text{mean}} \text{ 滿足一致性}$$

表示在樣本 $N \rightarrow \infty$ 的條件下, $\hat{B}_{2, \text{mean}}$ 為 B_2 的一致估計量
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(e)

```
+ #e小題
+ #i.
+ var_ols <- sigma2/sum((x - mean(x))^2)
+ var_mean <- (4*sigma2)/(N*(x2_bar-x1_bar)^2)
+ #ii.
+ sx2 <- sum((x-mean(x))^2)/N
+ term_i <- 1/sx2
+ term_ii = 4/(x2_bar - x1_bar)^2
+ results$var_b2_ols[i] <- var_ols
+ results$var_b2_mean[i] <- var_mean
+ results$estimated_sx2_inv[i] <- term_i
+ results$estimated_term2[i] <- term_ii
+ }
> print(results)
```

	N	var_b2_ols	var_b2_mean	estimated_sx2_inv	estimated_term2
1	100	1.15377235	1.59304290	0.1153772	0.1593043
2	500	0.24637040	0.33478244	0.1231852	0.1673912
3	1000	0.12344329	0.16625484	0.1234433	0.1662548
4	5000	0.02419076	0.03226408	0.1209538	0.1613204

$\text{Var}(b_2|x), \text{Var}(\hat{B}_{2, \text{mean}}|x), E[(s^2)^{-1}], E[4/(\bar{x}_2 - \bar{x}_1)^2]$ #

(f)

(i) 發現不管樣本數 N 為多少, $\text{Var}(b_2|x)$ 始終小於 $\text{Var}(\hat{B}_{2, \text{mean}}|x)$, 此符合

Gauss-Markov 定理, 即 OLS 估計量為 BLUE, 且隨著 N 愈大, 兩個變異數皆會趨近於 0, 說明 $\hat{B}_{2, \text{mean}}$ 亦為 B_2 之一致估計量。

(ii)

隨著 N 愈大, $E[(s^2)^{-1}]$ 會趨近於 0.12, $E[4/(\bar{x}_2 - \bar{x}_1)^2]$ 會趨近於 0.16

亦說明 OLS 估計量為最有效的不偏估計量。
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(g) $X \sim U(0, 10)$

$$\text{Var}(X) = \frac{(10-0)^2}{12} = \frac{100}{12}$$

$$E[(S^2)^{-1}] \xrightarrow{P} \frac{1}{\frac{100}{12}} = 0.12$$

對於經過排序後的均勻分配, $x_2 - x_1$ 之期望值差距大約為 5

$$\text{故 } E\left[\frac{4}{(\bar{x}_2 - \bar{x}_1)^2}\right] \xrightarrow{P} \frac{4}{25} = 0.16 \quad \#$$

(h)

若 X 無經過排序隨機分成兩組, 則 $E(\bar{x}_2) = E(\bar{x}_1) \Rightarrow$ 分母 $(\bar{x}_2 - \bar{x}_1) \approx 0$,
會導致估計量的 Variance 發散, 此時估計量不再符合一致性, 因為其
無法隨著樣本增加而提供穩定資訊。#